

Lecture 02 Assignments

Combinational Logic

1) Consider the following truth table (where A , B , and C are 3 input variables and Y is an output variable):

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

1.1) Write a Boolean equation in *sum-of-products* canonical form.

1.2) Minimize the equation from **1.1** using a ***K-map*** and sketch its combinational logic circuit.

2) Consider the following truth table (where A , B , and C are 3 input variables and Y is an output variable):

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

2.1) Write a Boolean equation in *sum-of-products* canonical form.

2.2) Minimize the equation from **2.1** using a ***K-map*** and sketch its combinational circuit.

3) Create the following logic functions using NAND gate(s):

3.1) a two-input AND

3.2) a two-input NOR

3.3) a two-input OR

3.4) a two-input XOR

4) Simplify the following logic equations and *sketch the circuits*:

$$4.1) Y = AB + \overline{A}BC + A\overline{B}C$$

$$4.2) Y = A + \overline{A}B + A\overline{B}C$$

$$4.3) Y = (\overline{A+B+C}) + \overline{A}B\overline{C} + A\overline{B}C$$

$$4.4) Y = (\overline{A+B})C + \overline{A}BC + ABC$$

Exercise 2.13 Simplify the following Boolean equations using Boolean theorems. Check for correctness using a truth table or K-map.

$$(a) Y = AC + \overline{A}\overline{B}C$$

$$(b) Y = \overline{A}\overline{B} + \overline{A}B\overline{C} + (\overline{A+C})$$

$$(c) Y = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C}\overline{D} + ABD + \overline{A}\overline{B}C\overline{D} + B\overline{C}D + \overline{A}$$

Exercise 2.14 Simplify the following Boolean equations using Boolean theorems. Check for correctness using a truth table or K-map.

$$(a) Y = \overline{A}BC + \overline{A}B\overline{C}$$

$$(b) Y = \overline{A}B\overline{C} + \overline{A}\overline{B}$$

$$(c) Y = ABC\overline{D} + \overline{A}B\overline{C}D + (\overline{A+B+C+D})$$

5) Create the circuit schematic of a **4-line Multiplexer** using its truth table.

6) Create a 4-bit *parity checker* using XOR gates.

7) Create the circuit schematic of a **4-line Decoder** using its truth table. Show the address ranges used by the decoder to select four memory chips (each with size 16K x 8).